

Evaluation of Grid-based Uncertainty Propagation for Collaborative Self-Calibration in Indoor Positioning Systems

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Abstract—Radio-based localization systems conventionally require stationary reference points (e.g. anchors) with precisely surveyed positions, making deployment time-consuming and costly. This paper presents an empirical evaluation of collaborative self-calibration for Ultra-Wideband (UWB) networks, extending a Bayesian approach based on grid-based uncertainty propagation. The enhanced algorithm reduces measurement availability requirements while maintaining positioning accuracy through probabilistic state estimation. We validate the approach using real-world data from controlled indoor experiments with 12 nodes in a static environment. Experimental evaluation yields 0.28 m mean ranging error under line-of-sight conditions and 1.11 m overall ranging error across mixed propagation scenarios. Results confirm the algorithm's resilience to measurement noise and partial connectivity scenarios typical in industrial deployments. We evaluate algorithm robustness under NLOS-contaminated initialization, showing graceful accuracy degradation (median error 0.62–0.99 m) compared to closed-form methods that exhibit substantial performance collapse (median error up to 2.43 m). The findings contribute to automated UWB network initialization for indoor positioning applications, reducing infrastructure dependency compared to manual anchor calibration procedures.

Index Terms—Auto-positioning, Markov Localization, Ultra-Wideband (UWB), Radio-based Localization, Wireless Sensor Network (WSN), Bayes Filter

I. INTRODUCTION

LOCATION-AWARE systems in intelligent transportation, smart cities, and logistics require sub-decimeter positioning accuracy for autonomous navigation, asset tracking, and human-machine interaction [1]. Traditional radio-based localization relies on stationary anchors with precisely surveyed positions, creating deployment complexity that scales superlinearly with network density. Manual anchor calibration represents a substantial percentage of system installation costs [2]. Collaborative networks intensify this challenge through systematic error propagation. Positioning errors create correlated uncertainties that compound throughout the system.

Collaborative self-calibration addresses these limitations by estimating node positions without a-priori knowledge of

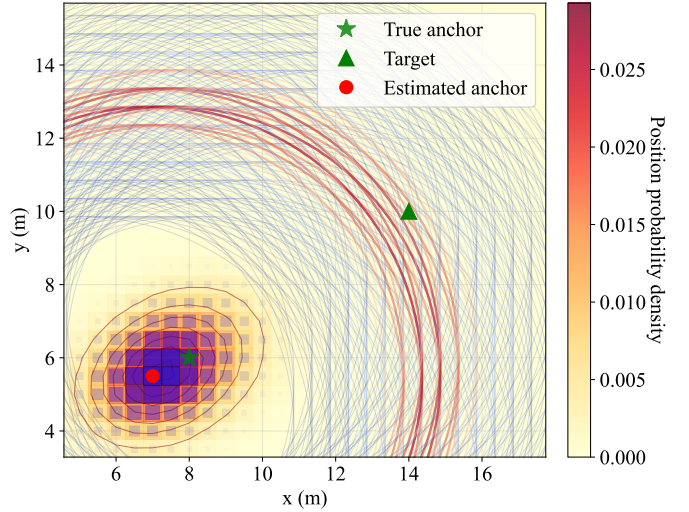


Fig. 1: Grid-based uncertainty propagation for collaborative self-calibration: Anchor probability distribution generates multiple weighted measurement hypotheses (radiating lines) rather than single point-estimate constraints, preserving spatial uncertainty information for reliable localization.

network topology [3]. However, parametric methods exhibit critical limitations when addressing non-Gaussian distributions and multimodal uncertainties prevalent in non-line-of-sight (NLOS) environments [2]. Recent surveys identify inadequate noise models and insufficient uncertainty quantification as fundamental failure modes [1], [4].

This research addresses collaborative self-calibration through probabilistic uncertainty propagation frameworks that achieve sub-meter positioning accuracy while reducing infrastructure dependencies. As illustrated in Fig. 1, the proposed grid-based approach generates multiple weighted measurement hypotheses rather than single point-estimate constraints, preserving spatial uncertainty information for robust target localization. The methodology extends discrete Bayesian estimation to maintain uncertainty distributions across network nodes, validated through experimental evaluation using real-world data from a fully meshed UWB sensor network under mixed propagation conditions.

This approach shows greater resilience to initialization parameters and measurement availability constraints compared

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to conventional parametric methods.

A. Related Work

Collaborative self-calibration encompasses automatic positioning procedures that enable anchor nodes to determine their locations without manual surveying, utilizing peer-to-peer ranging measurements [5], [6]. This approach differs from cooperative localization, which focuses on information sharing to improve individual node accuracy [7], by eliminating fixed infrastructure dependencies entirely. The fundamental challenge emerges from uncertainty propagation characteristics in collaborative networks, where positioning errors systematically bias dependent node estimates.

1) *Collaborative calibration*: methods enhance accuracy for anchors with approximate known coordinates. Qi et al. [8] address second-order error terms in trilateral localization through tightly coupled UWB-MEMS integration, demonstrating that anchor position errors significantly impact localization performance when neglected in conventional first-order approaches. Xu et al. [9] employ Modified Light Gradient Boosting Models with cooperative screening strategies to handle estimation errors across diverse network topologies.

2) *Meshed anchor-tag networks*: Technology-specific implementations demonstrate meshed approaches across wireless technologies. WiFi-RTT systems [10] utilize round-trip time measurements with SLAM approaches, while 5G-based systems [11] leverage combined range and angle-of-arrival measurements. UWB-aided vision navigation [12] and Calibration Unit approaches [13] provide alternative methodologies for inter-anchor distance estimation.

3) *Fully meshed networks*: achieve network-wide positioning without external references, representing the most autonomous positioning methodologies. Multidimensional Scaling (MDS) approaches form a foundational category, with Krapez et al. [14] demonstrating MDS integration achieving 0.32 m anchor localization accuracy in three-dimensional environments. Closed-form auto-positioning approaches [15], [16] provide computational efficiency through direct algebraic solutions, while machine learning enhanced methods [17] achieve centimeter-level accuracy through adaptive physical layer settings and signal correction algorithms.

However, existing parametric collaborative localization methods exhibit significant limitations when addressing non-Gaussian distributions and multimodal uncertainties prevalent in NLOS propagation environments [2]. Traditional approaches approximate node position uncertainty through covariance matrices and trace operations [18], effectively discarding spatial distribution structure. This parametric representation fails to capture true network-wide uncertainty propagation, motivating non-parametric approaches that preserve discrete probability mass distributions throughout collaborative positioning processes.

B. Scope and Contributions

This paper addresses the challenges of collaborative self-calibration in indoor positioning systems by evaluating a grid-based approach in a mixed LOS- and NLOS-environment,

where traditional parametric methods exhibit diminished performance due to NLOS propagation and measurement outliers. The primary contributions of this work are fourfold:

- 1) We present a self-surveyed dataset acquired from a fully meshed UWB sensor network employing two-way ranging (TWR), providing empirical validation data for ranging quality assessment that captures challenges inherent to NLOS reception and intermittent signal loss.
- 2) We extend and evaluate a previously introduced grid-based Bayesian formulation for collaborative self-calibration [19]. This approach employs non-parametric filtering that preserves uncertainty distributions across network nodes, showing greater resilience compared to parametric self-localization methods.
- 3) We show that the proposed method reduces connectivity requirements and measurement availability constraints compared to state-of-the-art approaches, enabling practical deployment in scenarios where conventional methods fail due to insufficient measurement density or network connectivity limitations.
- 4) We evaluate algorithm robustness under NLOS-contaminated anchor initialization. The grid-based method maintains sub-meter median accuracy while closed-form approaches exhibit order-of-magnitude degradation, confirming practical applicability in environments where LOS initialization cannot be guaranteed.

The remainder of this paper is structured as follows: Section II establishes theoretical foundations for collaborative self-calibration, introducing a baseline closed-form approach. Section III presents the proposed Bayesian grid-based approach with uncertainty propagation. Section IV describes the measurement campaign and UWB ranging data collection. Section V presents empirical results with comparative positioning performance analysis. The paper concludes with a synthesis of findings in Section VI.

II. FUNDAMENTALS

This section establishes the theoretical foundation for collaborative self-calibration by examining dependency structures in network-based positioning systems and presenting both baseline and proposed approaches. The analysis reveals limitations of conventional parametric uncertainty propagation methods, motivating the main contribution of this research.

A. Collaborative Positioning Dependencies

Collaborative positioning systems exhibit interdependencies that distinguish them from conventional anchor-based localization. Figure 2 illustrates the fundamental dependency structure, where $\mathbf{p}_k^{(i)}$ denotes the position vector of node i at time step k , $z_{ij}^{(k)}$ represents the distance measurement between nodes i and j , and $p(\cdot|\cdot)$ denotes conditional probability distributions.

Temporal evolution follows prediction models $p(\mathbf{p}_{k+1}^{(i)}|\mathbf{p}_k^{(i)})$ representing positional changes between time steps. The critical challenge emerges from spatial coupling: distance measurements $z_{ij}^{(k)}$ depend simultaneously on both node positions

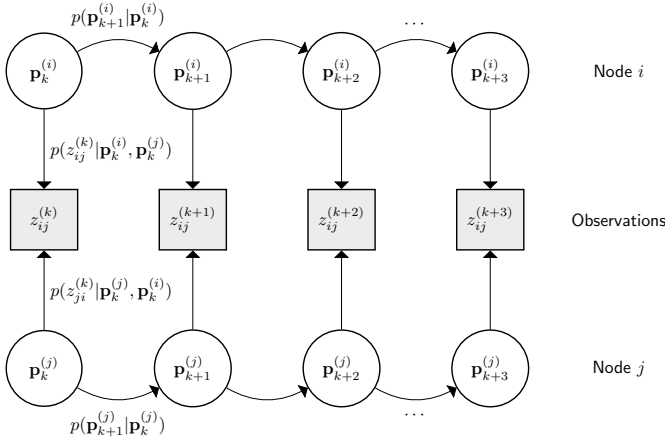


Fig. 2: Dependencies in collaborative positioning networks. Circles denote node states $\mathbf{p}_k^{(i)}$, squares represent observations $z_{ij}^{(k)}$, horizontal arrows indicate temporal evolution $p(\mathbf{p}_{k+1}^{(i)} | \mathbf{p}_k^{(i)})$, and vertical arrows represent measurement dependencies $p(z_{ij}^{(k)} | \mathbf{p}_k^{(i)}, \mathbf{p}_k^{(j)})$. This coupling structure creates correlated uncertainties that propagate throughout the network.

through $p(z_{ij}^{(k)} | \mathbf{p}_k^{(i)}, \mathbf{p}_k^{(j)})$. This bidirectional dependency creates systematic bias propagation where positioning errors in individual nodes affect estimates of dependent nodes.

Traditional parametric approaches approximate position uncertainties as additive noise terms, severing the dependency links in Figure 2. This approximation discards distributional information, particularly problematic for non-Gaussian or multimodal uncertainties common in NLOS environments. The resulting estimates become overconfident, failing to capture network-wide uncertainty propagation.

Collaborative systems require estimation methods that preserve and propagate uncertainty information across network nodes while maintaining computational tractability.

B. Baseline Closed-Form Approach

The baseline method, utilizing a fully meshed network as depicted in Fig. 3, establishes a coordinate reference frame using three anchors with the following assumptions [15], [16]:

- $\mathbf{a}_0$ is situated at the coordinate origin
- The direction from $\mathbf{a}_0$ to $\mathbf{a}_1$ defines the positive x-axis
- $\mathbf{a}_2$ lies in the half-plane with positive y-coordinate

The objective is estimating two-dimensional positions $\mathbf{p}_i = [x_i, y_i]^T$ for each node $i \in \{1, \dots, N\}$ using the inter-node distance matrix:

$$\mathbf{D}^{(k)} = \begin{pmatrix} z_{0,0}^{(k)} & z_{0,1}^{(k)} & \cdots & z_{0,N}^{(k)} \\ z_{1,0}^{(k)} & z_{1,1}^{(k)} & \cdots & z_{1,N}^{(k)} \\ \vdots & \vdots & \ddots & \vdots \\ z_{N,0}^{(k)} & z_{N,1}^{(k)} & \cdots & z_{N,N}^{(k)} \end{pmatrix} \quad (1)$$

where $z_{ij}^{(k)}$ represents measured distance between nodes i and j at time step k . We distinguish between true geometric distances $d_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|_2$ and measured distances $z_{ij}^{(k)} = d_{ij} + n_{ij}^{(k)}$, where $n_{ij}^{(k)}$ represents measurement error.

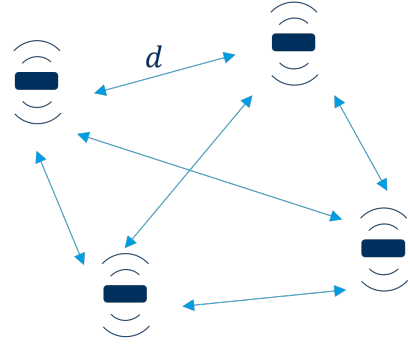


Fig. 3: Fully meshed sensor network configuration

The distance matrix is populated through direct pairwise ranging measurements between all node combinations. Our fully meshed architecture performs direct two-way ranging (TWR) between each node pair without intermediate relay nodes, distinguishing this approach from multi-hop collaborative localization methods [2]. While the experimental deployment provides a fully meshed network, the proposed grid-based method does not require a complete distance matrix. The probabilistic framework accommodates partial connectivity through selective reference anchor utilization in the measurement update, as demonstrated in Section V-B.

The coordinate frame is established with $\mathbf{a}_0 = [0, 0]^T$ and $\mathbf{a}_1 = [z_{01}^{(k)}, 0]^T$. Subsequent nodes are positioned by:

$$\mathbf{a}_2 = \left[\frac{(z_{02}^{(k)})^2 - (z_{12}^{(k)})^2 + x_1^2}{2x_1}, \sqrt{(z_{02}^{(k)})^2 - x_2^2} \right]^T. \quad (2)$$

This approach exhibits several limitations that motivate the probabilistic framework presented in Section III: (1) measurement errors propagate algebraically through squared terms without uncertainty quantification, (2) measurement failures or high ranging errors render trilateration infeasible due to geometric inconsistencies, (3) the square root operation introduces numerical instabilities when $d_{02}^2 < x_2^2$, and (4) the deterministic coordinate frame depends critically on the first three anchor positions, creating systematic sensitivity to initialization selection as demonstrated in Section V-C.

III. BAYESIAN GRID-BASED APPROACH

The key methodological innovation extends the grid-based framework of [19] by implementing hypothesis-based uncertainty propagation rather than parametric approximation of anchor position uncertainties. While parametric approaches collapse probability distributions to covariance matrices, the proposed method preserves and propagates discrete probability mass from anchors, enabling more accurate uncertainty quantification in collaborative positioning scenarios.

This is achieved by incorporating weighted contributions from multiple position hypotheses into the likelihood calculation as part of the correction step in a recursive state estimation framework. The overall procedure is summarized in Fig. 4 and detailed in the following.

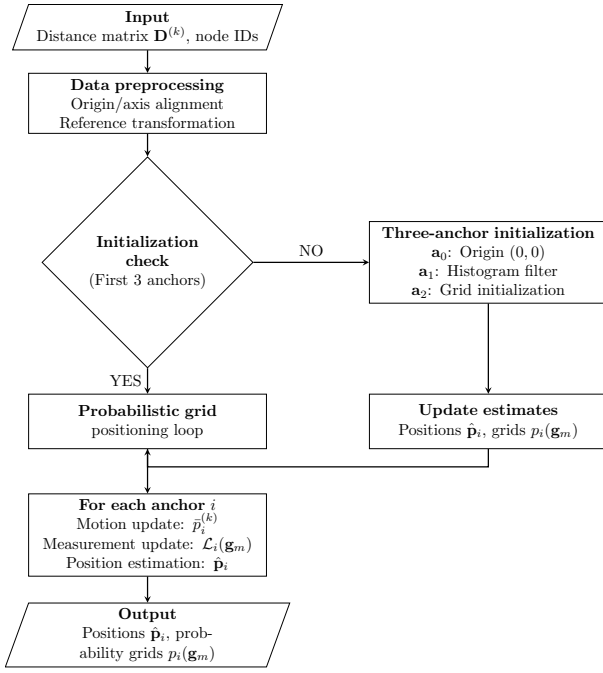


Fig. 4: Algorithmic flowchart of the grid-based self-calibration method. The process implements distance matrix preprocessing, sequential three-anchor initialization ($\mathbf{a}_0$ at origin, $\mathbf{a}_1$ via histogram-, $\mathbf{a}_2$ via grid filter), and recursive Bayesian filtering with uncertainty propagation.

Discrete grid representation of the state space enables non-parametric probability density estimation that is resilient to non-Gaussian measurement noise and multimodal distributions. For each node, the continuous position space is discretized into M equidistant grid points $\mathcal{G} = \{\mathbf{g}_1, \mathbf{g}_2, \dots, \mathbf{g}_M\}$.

Posterior probabilities over the grid are maintained and updated following Bayes' rule:

$$\text{posterior} \propto \text{likelihood} \times \text{prior} \quad (3)$$

1) *Parameter Selection*: The grid-based approach employs the following parameters:

- M – Total number of discretized grid cells
- τ – Percentage threshold for probability mass selection
- r_{est} – Estimation region radius
- σ_r – Ranging measurement standard deviation
- v – Expected node velocity
- σ_v – Velocity uncertainty standard deviation

Table I summarizes parameter values and selection rationale. Grid resolution balances spatial accuracy against computational cost, whereas the probability threshold τ captures primary hypotheses while limiting computation. Ranging noise σ_r derives from LOS statistics (Table II) with safety margin.

A. Initialization Phase

The proposed method leverages the coordinate frame established in Section II-B while implementing probabilistic estimation. Unlike the deterministic closed-form approach, the initialization phase maintains uncertainty distributions throughout coordinate system establishment.

TABLE I: Grid-based method parameters.

Parameter	Value	Rationale
M	$\sim 13,200$	0.1 m spacing over $44 \times 30 \text{ m}^2$
τ	3%	Primary hypotheses
r_{est}	2 m	Local averaging without over-smoothing
σ_r	0.5 m	LOS statistics + margin
σ_v	0.1 m	Filter recovery in static scenarios

1) *Anchor 0 - Origin Definition*: Following the baseline methodology, the first anchor defines the coordinate system origin $\mathbf{a}_0 = [0, 0]^T$. No uncertainty is associated with this mandatory definition.

2) *Anchor 1 - Histogram Filter*: The second anchor position is estimated along the positive x-axis using a one-dimensional histogram filter with H hypotheses over the range $[0, d_{\text{max}}]$:

$$p_1(h_j) = \frac{\exp\left(-\frac{(z_{01}^{(k)} - h_j)^2}{2\sigma_r^2}\right)}{\sqrt{2\pi\sigma_r^2}}, \quad (4)$$

where h_j represent the histogram bins, σ_r denotes the ranging measurement standard deviation, and d_{max} represents the maximum operational range. The position estimation reduces to a one-dimensional problem while maintaining probabilistic uncertainty representation. The non-parametric nature of the histogram filter enables incorporation of arbitrary probability distributions without restrictive Gaussian assumptions, facilitating stable estimation under multimodal or heavy-tailed measurement error conditions.

3) *Anchor 2 - Grid Initialization*: The third anchor represents the first implementation of the grid-based uncertainty propagation framework, establishing the foundational probabilistic estimation method applied to all subsequent anchors. Unlike the deterministic origin definition $\mathbf{a}_0$ and one-dimensional histogram approach $\mathbf{a}_1$, $\mathbf{a}_2$ employs full two-dimensional Bayesian grid estimation. The initialization procedure utilizes three inter-node distance measurements $(z_{01}^{(k)}, z_{02}^{(k)}, z_{12}^{(k)})$ within the grid-based measurement update framework. The trilateration constraints from Section II-B provide geometric initialization of the applicable state space. The grid-based uncertainty propagation processes the three distance measurements through the complete Bayesian framework:

$$p_2(\mathbf{g}_m) = \eta \cdot \prod_{j \in \{0,1\}} \exp\left(-\frac{(\|\mathbf{g}_m - \mathbf{a}_j\| - z_{j2}^{(k)})^2}{2\sigma_r^2}\right) \quad (5)$$

where η is the normalization constant ensuring $\sum_m p(\mathbf{g}_m) = 1$ and $\mathbf{a}_j$ denotes the established positions of anchors 0 and 1. The subsequent position estimation $\hat{\mathbf{a}}_2$ follows the approach detailed in Section III-B.3.

4) *Initialization Robustness*: The probabilistic initialization provides inherent robustness to NLOS-contaminated measurements compared to closed-form trilateration. Rather than computing a single point solution that fails under geometric inconsistency, Equation (5) weights grid cells by measurement consistency across all three initialization distances. NLOS bias

in one measurement creates probability mass in regions potentially inconsistent with the remaining measurements, resulting in broader but more representative uncertainty distributions.

To evaluate initialization robustness, we define two additional configurations (Conf. 3 and Conf. 4) where 2 of 3 initial anchor links are NLOS-contaminated (see Section V-C). The grid-based method achieves median errors of 0.62–0.99 m under NLOS initialization compared to 0.34–0.68 m under LOS, demonstrating graceful degradation. In contrast, the closed-form approach exhibits median errors of 1.48–2.43 m with extreme outliers exceeding 30 m, confirming the robustness advantage of probabilistic uncertainty propagation.

B. Recursive Position Estimation

After initialization, the remaining anchors employ a recursive Bayesian filter structure (Fig. 5), alternating between prediction and update steps.

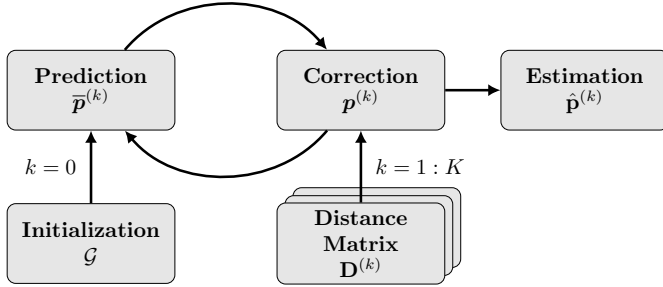


Fig. 5: Recursive filter structure for grid-based self-calibration.

1) *Motion update*: The prediction step incorporates expected node movement:

$$\bar{p}_i^{(k)} = \sum_{m \in \mathcal{K}} p_m^{(k-1)} \cdot p(\mathbf{g}_i^{(k)} | \mathbf{g}_m^{(k-1)}) \quad (6)$$

where $\mathcal{K}$ represents the set of grid cells with the highest probabilities. This selection reduces computational complexity while maintaining estimation accuracy. The transition probability is modeled as:

$$p(\mathbf{g}_i^{(k)} | \mathbf{g}_m^{(k-1)}) = \mathcal{N}(\|\mathbf{g}_i - \mathbf{g}_m\|_2 - v\Delta t, \sigma_v^2) \quad (7)$$

with velocity v and velocity uncertainty σ_v . For static scenarios, $v = 0$ but $\sigma_v > 0$ adds noise to enable filter recovery from erroneous estimates.

2) *Measurement update with uncertainty propagation*: The core contribution of this measurement update addresses uncertainty propagation in collaborative positioning through discrete probability mass preservation rather than parametric approximation. Traditional approaches approximate reference position uncertainty through covariance matrices and apply trace operations [18]:

$$\sigma_{1,2} = \sigma_r + \sqrt{\text{tr}(\mathbf{P}_1)} + \sqrt{\text{tr}(\mathbf{P}_2)} \quad (8)$$

to compute aggregate measurement noise. This parametric representation discards spatial distribution, particularly problematic when positions exhibit multimodal or non-Gaussian distributions common in NLOS-contaminated environments.

The proposed hypotheses-based approach preserves distributional information by maintaining discrete probability mass representations. Rather than collapsing anchor j 's position distribution to a single covariance matrix, the method samples the top $\tau\%$ probability mass as discrete hypotheses:

$$\mathcal{T}_j^\tau = \{\mathbf{g}_{jl} : l \in \arg \max_m p_j(\mathbf{g}_m)\} \quad (9)$$

where each hypothesis retains its associated probability weight $w_{jl} = p_j(\mathbf{g}_{jl}) / \sum_{k \in \mathcal{T}_j^\tau} p_j(\mathbf{g}_{jk})$.

For anchor i being updated, the measurement likelihood incorporates uncertainty from all anchors $j \in \mathcal{N}_i$ through weighted contributions from multiple position hypotheses:

$$\mathcal{L}_i(\mathbf{g}_m) = \prod_{j \in \mathcal{N}_i} \sum_{l \in \mathcal{T}_j^\tau} w_{jl} \cdot \exp \left(-\frac{(\|\mathbf{g}_m - \mathbf{g}_{jl}\|_2 - z_{ij}^{(k)})^2}{2\sigma_r^2} \right) \quad (10)$$

where $\mathcal{N}_i$ is the set of available reference anchors for node i , $\mathcal{T}_j^\tau$ represents the top $\tau\%$ probability cells of anchor j , w_{jl} are the normalized weights for each hypothesis and $z_{ij}^{(k)}$ is the measured distance between nodes i and j at time step k .

This weighted superposition captures the true uncertainty propagation from anchors to target positions, as illustrated in Fig. 6. The parametric approximation generates circular measurement constraints from point estimates, while the hypotheses-based method produces aggregate likelihood fields that reflect the actual spatial uncertainty distribution of anchor positions. This distinction becomes increasingly important as positioning errors accumulate through the network.

The posterior probability is updated using Bayes' rule with normalization constant η :

$$p_i(\mathbf{g}_m) = \eta \cdot \bar{p}_i(\mathbf{g}_m) \cdot \mathcal{L}_i(\mathbf{g}_m) \quad (11)$$

3) *Position estimation*: The position estimate at each time step is computed as a weighted average in the circular region of radius r around the maximum likelihood estimate with $\mathcal{B}_r(\mathbf{g}_{\text{MLE}}) = \{m : \|\mathbf{g}_m - \mathbf{g}_{\text{MLE}}\|_2 < r\}$:

$$\hat{p}_i = \frac{\sum_{m \in \mathcal{B}_r(\mathbf{g}_{\text{MLE}})} p_i(\mathbf{g}_m) \cdot \mathbf{g}_m}{\sum_{m \in \mathcal{B}_r(\mathbf{g}_{\text{MLE}})} p_i(\mathbf{g}_m)} \quad (12)$$

4) *Computational complexity*: The per-epoch complexity scales as $\mathcal{O}(\tau^2 M^2 N^2)$. For our configuration with $M \approx 13,200$ grid cells, $N = 12$ nodes, and probability threshold $\tau = 0.03$, this yields approximately 2.3×10^7 likelihood evaluations per network update. The centralized architecture aggregates all $N(N-1)/2 = 66$ pairwise measurements at a single coordinator. Future distributed implementations could reduce per-node computation to $\mathcal{O}(\tau^2 M^2 |\mathcal{N}_i|)$ by restricting likelihood updates to local neighbor sets.

IV. EXPERIMENTAL SETUP AND DATA COLLECTION

To validate the proposed grid-based positioning method, we conducted controlled experiments in a real-world industrial environment within a business park in Torgau, Germany. The experimental campaign utilized a fully meshed UWB sensor network enabling comprehensive inter-node ranging measurements under mixed propagation conditions.

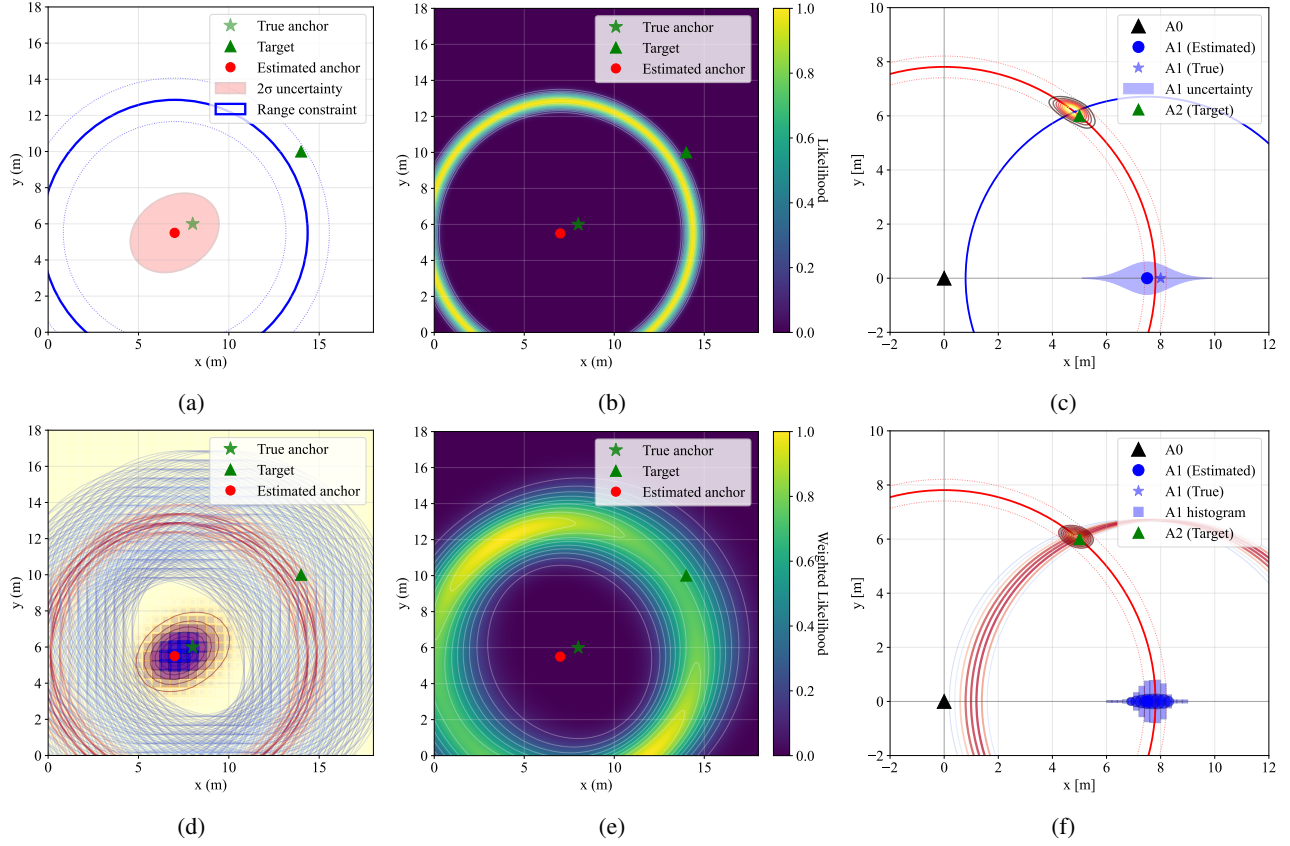


Fig. 6: Uncertainty propagation for collaborative self-calibration: Top row presents traditional point-estimate approach; bottom row demonstrates the proposed grid-based method. (a) Traditional parametric representation of reference anchor as point estimate with associated covariance. (d) Grid-based representation showing probability distribution (colormap), sampled position hypotheses (colored points), and measurement constraints (circles). (b, e) Likelihood computation: (b) single measurement constraint from point estimate; (e) weighted contributions from multiple position hypotheses demonstrating aggregate likelihood fields. (c) Parametric initialization procedure. (f) Grid-based initialization using hypotheses-based uncertainty propagation.

A. Hardware Configuration

The experimental setup employed 12 Qorvo MDEK1001 UWB transceivers operating in the 3.5-6.5 GHz frequency band. The original firmware was customized by Deveritec GmbH under proprietary licensing to enable full mesh networking capabilities, supporting peer-to-peer ranging between all node pairs through direct two-way ranging (TWR) without intermediate relay hops. Architectural descriptions of similar fully meshed UWB implementations using the same hardware platform are available in [20].

One node serves as network coordinator, responsible for orchestrating the sequential ranging operations to prevent channel conflicts through time-division multiplexing. The network operated with 10 Hz measurement frequency, representing the aggregate system update rate after all pairwise ranging operations complete sequentially within each epoch. The centralized coordinator is also responsible for data aggregation.

B. Deployment configuration and reference system

We deployed a network of 12 nodes throughout a 44×30 m² indoor hall (see Fig. 8), mounted on tripods at 1.60 m height. Node placement creates diverse propagation scenarios,

including both LOS and NLOS configurations essential for robustness evaluation under realistic deployment conditions.

Nodes were powered using USB mains power supplies or USB battery packs, enabling flexible placement. The sensor network was configured and controlled via PC interface, providing access to sensor data and network parameters.

We documented visibility conditions between node pairs, creating a binary LOS/NLOS classification matrix for subsequent propagation-dependent analysis. The resulting propagation conditions encompass moderate NLOS scenarios typical of industrial hall environments, characterized by partial occlusions from equipment and architectural elements rather than complete signal blockage.

We established reference coordinates using an optical Leica TS16 total station. A surveying prism served as the target for tachymeter measurements (see Fig. 8). The total station recorded precise coordinates of the prism, enabling creation of a local coordinate system based on predefined reference points in the environment.

For compatibility with the proposed algorithm, we applied a coordinate transformation. Two selected nodes define the positive x-axis, with the first node at the coordinate system origin, consistent with the assumptions in Section II-B. Figure 7

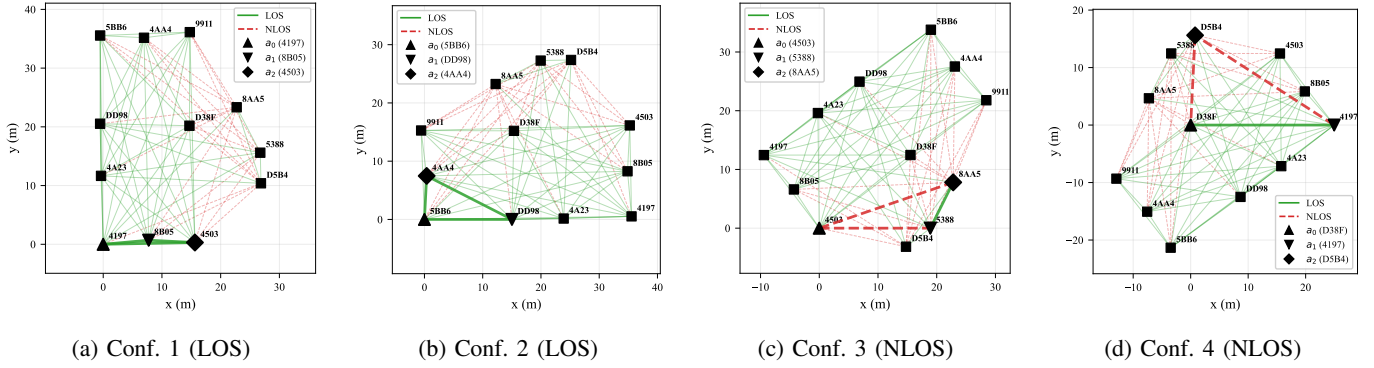


Fig. 7: Node constellations and visibility conditions for four initialization configurations. Green solid lines indicate LOS links, red dashed lines indicate NLOS links. Thick lines highlight the initialization triangle ($\mathbf{a}_0$ – $\mathbf{a}_1$ – $\mathbf{a}_2$). (a, b) Conf. 1 and 2: pure LOS initialization. (c, d) Conf. 3 and 4: NLOS-contaminated initialization with 2/3 NLOS links in the initialization triangle.



Fig. 8: Measurement environment with distributed nodes (red) and total station setup for ground truth position surveying.

illustrates the resulting four node configurations: two with pure LOS initialization and two with NLOS-contaminated initialization. This design enables systematic evaluation of anchor initialization sensitivity under varying propagation conditions.

C. Dataset Characteristics

The experimental campaign generated approximately 132,000 individual ranging measurements across all node pairs over 2,000 measurement epochs. Each epoch captured the complete 12×12 distance matrix with systematic documentation of ranging availability and residuals. Data are structured in text format with event timestamps, node identifiers, symmetric distance matrices, ground truth coordinates, and binary visibility condition matrices. Four anchor initialization configurations (Fig. 7), two with pure LOS and two with NLOS-contaminated initialization, enable systematic evaluation of algorithm sensitivity to reference frame selection and initialization propagation conditions. This supports reproducible performance assessment across varying measurement availability and propagation scenarios.

V. VALIDATION AND RESULTS

This section validates the proposed grid-based positioning method by analyzing ranging performance characteristics and comparing positioning accuracy between the baseline closed-form (CF) approach and probabilistic grid-based positioning (PGP) across four anchor initialization configurations.

A. Ranging Performance Analysis

The ranging performance evaluation utilizes the complete dataset of 132,000 individual measurements from 12 UWB nodes deployed according to Fig. 7. Statistical analysis is categorized by propagation conditions (LOS/NLOS) to quantify measurement quality variations that directly impact collaborative positioning performance.

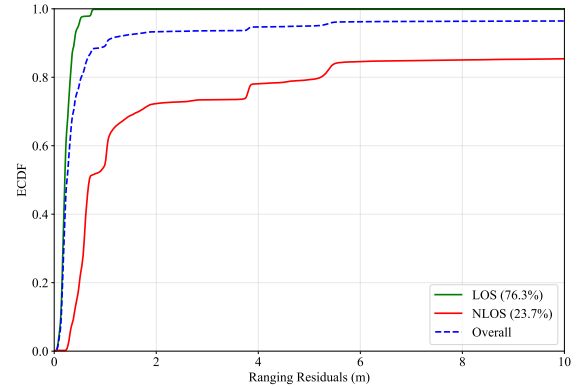


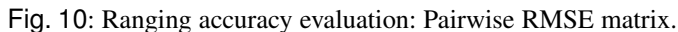
Fig. 9: Ranging residual ECDF (LOS, NLOS, overall).

TABLE II: Ranging performance statistics.

Metric	LOS (76.3 %)	NLOS (23.7 %)	Overall (100 %)
Mean μ (m)	0.28	3.78	1.11
Median (m)	0.21	0.69	0.25
Variance σ^2 (m)	1.59	41.56	13.26
25th	0.16	0.54	0.18
50th	0.21	0.69	0.25
75th	0.29	3.79	0.44

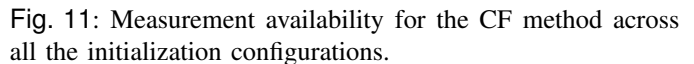
Table II presents ranging accuracy statistics across propagation conditions. LOS measurements achieve RMSE of 0.28 m with 0.21 m median error, exhibiting near-Gaussian characteristics. NLOS conditions show degraded performance with RMSE of 3.78 m, reflecting multipath propagation and signal obstruction. Across both conditions, overall RMSE is 1.11 m with variance $\sigma^2 = 13.26 \text{ m}^2$, indicating substantial measurement heterogeneity. The moderate NLOS error

The empirical cumulative distribution function (Fig. 9) provides statistical characterization aligned with IPIN competition evaluation standards [21], where the 75th percentile serves as the primary benchmark. LOS measurements exhibit tight error distribution (75th percentile: 0.29 m, 95th: 0.73 m), while NLOS exhibits heavy-tailed distribution with 75th percentile at 3.79 m. Still, 50% of NLOS measurements remain below 0.69 m, preserving ranging utility under moderate occlusion. The overall 75th percentile (0.44 m) informed the grid-based algorithm’s ranging noise parameter ($\sigma_r = 0.5$ m). The pairwise RMSE matrix (Fig. 10) reveals systematic quality variations: nodes D5B4, 8AA5, and 9911 exhibit elevated error rates exceeding 17m RMSE, correlating with NLOS-dominated positions (Fig. 7).



The availability analysis (Fig. 11) quantifies CF method sensitivity to anchor selection across all four configurations. Sensor D5B4 achieves zero positioning availability in both Conf. 1 and Conf. 3 due to missing ranges to initial anchors (Fig. 10), yet recovers to 55.6% and 47.0% in Conf. 2 and Conf. 4, respectively. Similarly, sensors 4197 and 8AA5 drop from 100.0% and 74.2% in Conf. 1 to 3.0% each in Conf. 2. The NLOS-contaminated configurations expose additional vulnerabilities: in Conf. 3, sensors 4AA4 and 9911 collapse to 0.4% and 9.6% respectively, despite maintaining $\approx 75\%$ availability across all other configurations.

These variations result from geometric constraints when insufficient connectivity occurs between the reference frame and target nodes. Cross-referencing with Fig. 10 reveals that positioning failures do not correspond to measurement quality degradation. Node D5B4 maintains ranging capabilities with



The requirement for partial rather than complete distance matrices represents a fundamental methodological advantage for deployments with heterogeneous measurement availability.

Positioning performance evaluation compares CF against PGP using identical measurement datasets across four initialization configurations (Fig. 7): two with pure LOS initialization (Conf. 1 and 2) and two with NLOS-contaminated initialization (Conf. 3 and 4), where 2 of 3 initial anchor links are NLOS-affected. This design isolates the effect of initialization propagation conditions on positioning accuracy. Summary statistics are provided in Table III, with ECDFs shown in Figs. 12 and 13.

1) *LOS Initialization Baseline*: Under pure LOS initialization, both methods achieve competitive performance. PGP attains median errors of 0.34–0.68 m across the two LOS configurations, with 66–92% of estimates below 1 m (Fig. 12). The CF method achieves comparable median errors (0.40–0.47 m) under these favorable conditions, though its 95th percentile for Conf. 1 (7.59 m) already reveals vulnerability to outliers that the PGP method avoids (2.00 m). The configuration-

TABLE III: Positioning error summary across four initialization configurations.

Config	Method	Median (m)	p75 (m)	p95 (m)	<1 m (%)	Mean (m)
Conf. 1 (LOS)	PGP	0.68	1.14	2.00	66.1	0.79
Conf. 1 (LOS)	CF	0.47	0.83	7.59	77.2	1.28
Conf. 2 (LOS)	PGP	0.34	0.53	1.12	91.6	0.42
Conf. 2 (LOS)	CF	0.40	0.66	1.84	79.4	0.67
Conf. 3 (NLOS)	PGP	0.62	1.04	2.05	73.3	0.76
Conf. 3 (NLOS)	CF	2.43	3.82	7.18	29.9	2.84
Conf. 4 (NLOS)	PGP	0.99	1.66	2.54	50.3	1.10
Conf. 4 (NLOS)	CF	1.48	17.77	30.35	45.2	8.14

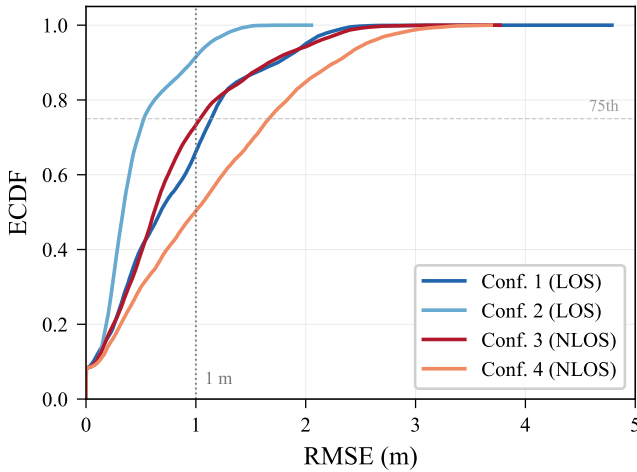


Fig. 12: ECDF of positioning error for the PGP method across four initialization configurations. Vertical dotted line: 1 m threshold. Horizontal dashed line: 75th percentile.

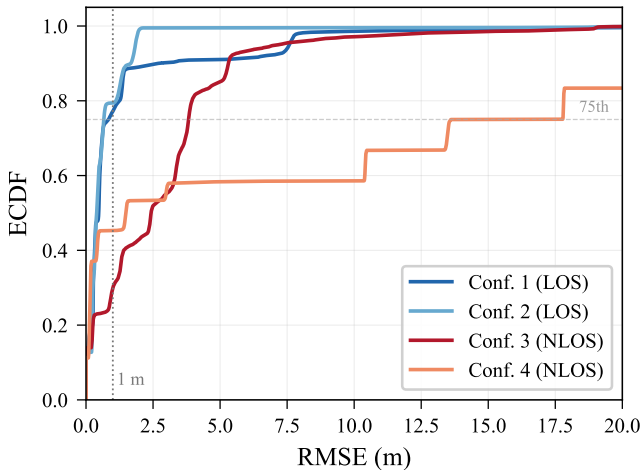


Fig. 13: ECDF of positioning error for the CF method across four initialization configurations. Note the extended x-axis range (0–20 m) compared to PGP (0–5 m).

dependent variation in both methods reflects geometric dilution of precision from different anchor placements.

2) *NLOS-Contaminated Initialization*: The critical distinction emerges under NLOS-contaminated initialization (Table III, lower half). The PGP method exhibits graceful degradation: median errors increase to 0.62–0.99 m (from 0.34–0.68 m under LOS), with the 75th percentile remaining below 1.66 m. Notably, PGP under Conf. 3 (median 0.62 m) performs comparably to PGP under Conf. 1 (median 0.68 m), indicating that moderate NLOS contamination does not necessarily prevent accurate positioning.

In contrast, the CF method exhibits substantial performance collapse under NLOS initialization. Median errors increase to 1.48–2.43 m, the 75th percentile reaches 17.77 m for Conf. 4, and extreme outliers exceed 30 m at the 95th percentile (Fig. 13). The fraction of sub-meter estimates drops from 77–79% (LOS) to 30–45% (NLOS). This degradation reflects the deterministic nature of closed-form trilateration: NLOS bias in the initialization triangle propagates directly into the coordinate frame definition, systematically displacing all subsequent position estimates.

3) *Robustness Analysis*: The PGP–CF performance gap widens substantially under NLOS initialization. While both methods achieve similar median accuracy under LOS conditions, the mean error ratio (CF/PGP) increases from $1.6\times$ under LOS to $3.7\text{--}7.4\times$ under NLOS. This robustness stems from the grid-based method’s probabilistic uncertainty propagation: NLOS-biased measurements create broader but representative probability distributions rather than catastrophic point failures. The measurement update (Eq. (10)) weights grid cells by consistency across all available references, enabling the posterior to recover from initialization bias as additional measurements accumulate.

VI. CONCLUSION

This paper presented grid-based uncertainty propagation for collaborative UWB self-calibration, addressing limitations of parametric methods that discard spatial distribution information when reducing anchor positions to point estimates. The proposed discrete Bayesian approach preserves probability mass throughout positioning, enabling reliable uncertainty quantification in NLOS-contaminated environments.

Experimental validation with 12 UWB nodes in a $44\times 30\text{ m}^2$ industrial environment generated 132,000 ranging measurements. LOS conditions achieved 0.28 m mean ranging error versus 3.78 m for NLOS. The grid-based approach achieved consistent sub-meter positioning accuracy under LOS initialization, while baseline closed-form methods exhibited substantial sensitivity to initialization selection.

Under NLOS-contaminated initialization, the grid-based approach maintained sub-meter median accuracy (0.62–0.99 m) while the closed-form baseline exhibited order-of-magnitude degradation with median errors up to 2.43 m and extreme outliers exceeding 30 m, confirming the practical advantage of probabilistic uncertainty propagation for deployments where LOS initialization cannot be guaranteed.

Future research directions include quantitative comparison with MDS and robust localization baselines, as well as dis-

tributed processing architectures for large-scale deployments. Multi-technology integration combining WiFi, BLE, and 5G within the probabilistic framework could enhance coverage. Machine learning may further enable automated measurement quality assessment.

DATA AVAILABILITY STATEMENT

The presented dataset, comprising 132,000 rangings with ground truth, LOS/NLOS classifications, and node configurations, will be made publicly available through the institutional repository of TU Dresden University of Technology following publication, facilitating reproducibility and future research.

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